

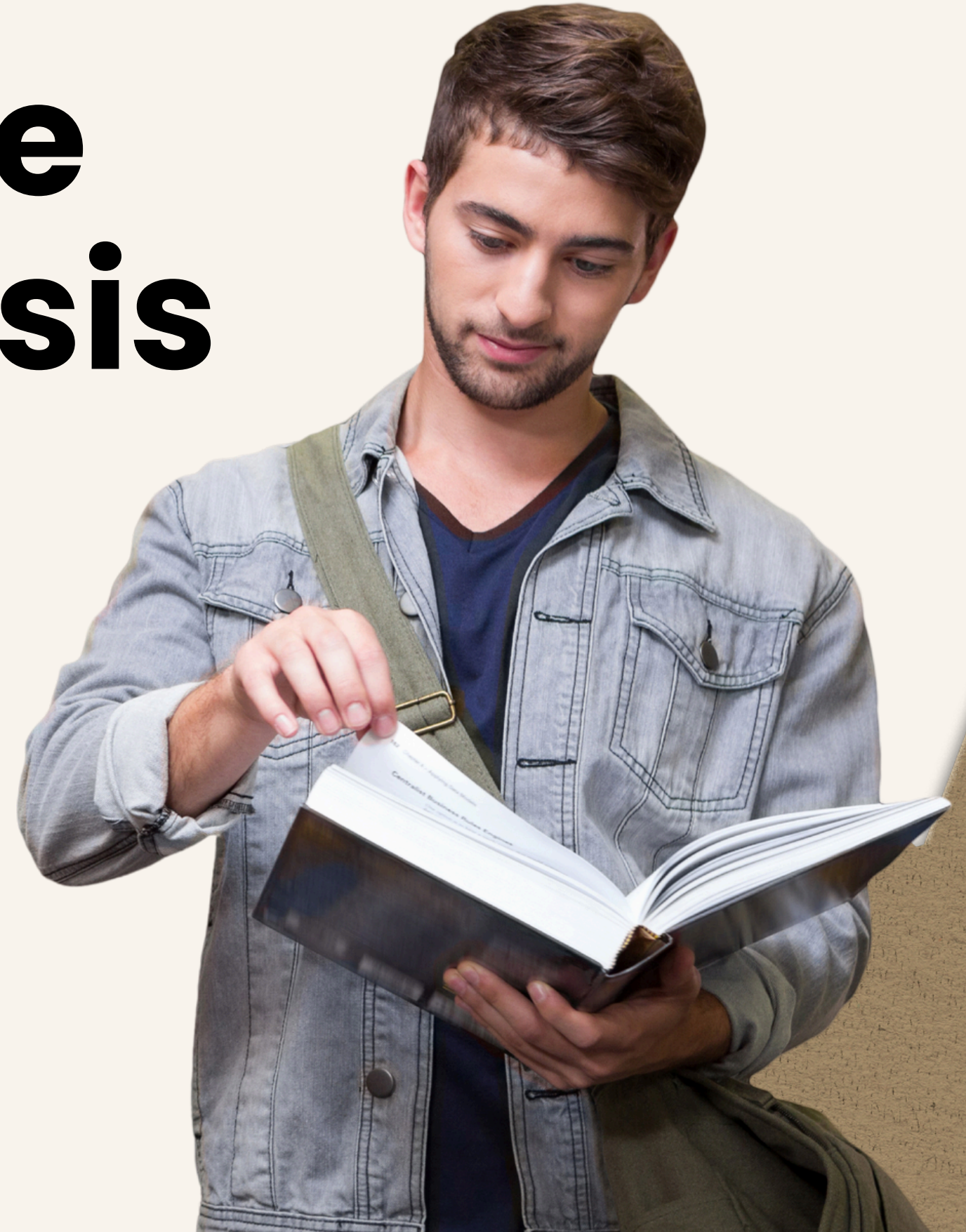
Natural Language Processing



Emotion Aware Journal Analysis System



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Introduction



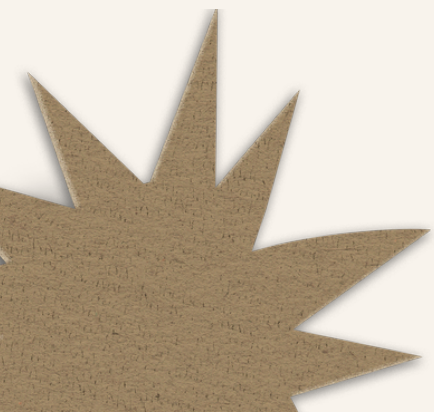
In today's fast-paced digital world, many individuals struggle to manage their emotional well-being before it leads to burnout or chronic stress. While personal journaling generates valuable emotional data, it often remains unanalyzed due to the difficulty of identifying patterns in unstructured text. Existing tools rely on basic sentiment analysis, which fails to capture emotional complexity and changes over time. This project proposes an NLP-based system that analyzes written and spoken journal entries to detect emotional patterns, provide personalized wellness recommendations, and adapt workload to prevent burnout while ensuring user privacy.

Problem Statement



Personal journaling is a common self-reflection practice, but its unstructured nature makes it difficult to identify emotional patterns, stress buildup, or long-term trends. As a result, early signs of burnout often go unnoticed, and existing tools using basic sentiment analysis fail to capture emotional complexity over time.

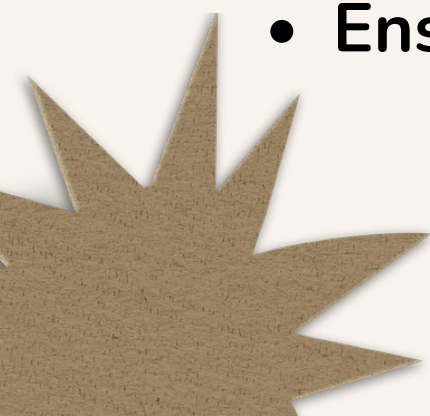
This project addresses the gap by proposing a privacy-aware NLP-based system that analyzes journal entries to detect emotional patterns and provide meaningful, wellness-focused insights, while ensuring user privacy and avoiding medical diagnosis.



Objectives



- Develop an NLP-based system to extract fine-grained emotions from user journal entries (text + voice)
- Enable multimodal emotion detection by combining textual analysis with speech-based emotional cues
- Perform longitudinal emotional trend analysis across multiple journal entries to detect patterns over time
- Design a mechanism to compute burnout risk scores using emotional intensity, frequency, and duration
- Generate personalized wellness recommendations based on user interests and detected emotional states
- Implement a closed-loop adaptive workload system that suggests task prioritization and rest intervals
- Ensure a privacy-preserving architecture by limiting processing to derived emotional features only

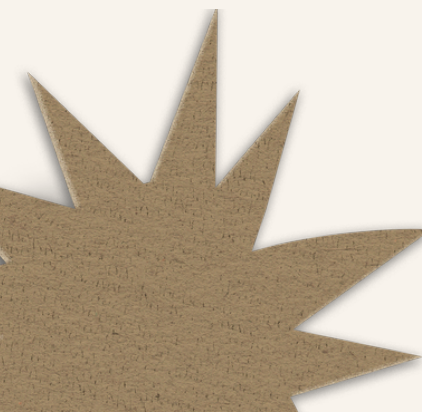


Literature Review

Sl No	Author & Year	Dataset	Methodology	Limitation
1	LER Model, 2025	ISEAR	RoBERTa + LSTM hybrid	No journal data, high compute
2	Multimodal Emotion AI, 2025	Text + Audio	RoBERTa + DistilBERT + speech features	Weak fusion, no noise handling
3	Kurpicz-Briki et al., 2025	YouTube + Synthetic	RuBERT burnout detection	Language + domain limitation
4	Emotion-Adaptive Music, 2025	EmoBank + Spotify	Transformer + Wide & Deep	Poor long-text generalization
5	AI Burnout Monitoring, 2025	Work logs	ML on behavioral patterns	Privacy concerns, domain-specific
6	ERC Survey, 2025	Multiple datasets	Review of DL models (LSTM, Transformer)	No journal focus

Sl No	Author & Year	Dataset	Methodology	Limitation
7	Emotion Drift Study, 2025	Reddit + Clinical	Sentence-level transformer + drift tracking	No multi-entry tracking
8	Multimodal Review, 2025	Multiple benchmarks	Fusion strategies review	No new model
9	Passive Stress Detection, 2025	Biometric + logs	Wearables + AI	Not NLP-focused
10	Emotion-based Music Rec., 2025	Emotion + music logs	Regression + Markov model	Subjective data bias
11	AI Wellbeing Framework, 2024	Behavioral data	Data-centric ML framework	Mostly conceptual
12	MER Survey, 2025	Multiple datasets	NLP + audio + metadata fusion	No journal input

Sl No	Author & Year	Dataset	Methodology	Limitation
13	Transformer Emotion Detection, 2025	Mental health text	BERT, RoBERTa benchmarking	Poor reproducibility
14	SER Review, 2025	Clinical speech	Deep learning on voice signals	No real-world validation
15	Burnout NLP (GermanBERT), 2024	Real + synthetic text	Domain transfer evaluation	Language + dataset limits
16	Attention-based Music Rec., 2025	Music datasets	CNN + RNN + Attention	No text/voice input
17	Human-centered AI, 2025	Survey (320 users)	Statistical burnout analysis	Correlational only
18	Emotion Detection Review, 2024	330 papers	ML + DL survey	No multimodal focus



◆ **Research Gap**

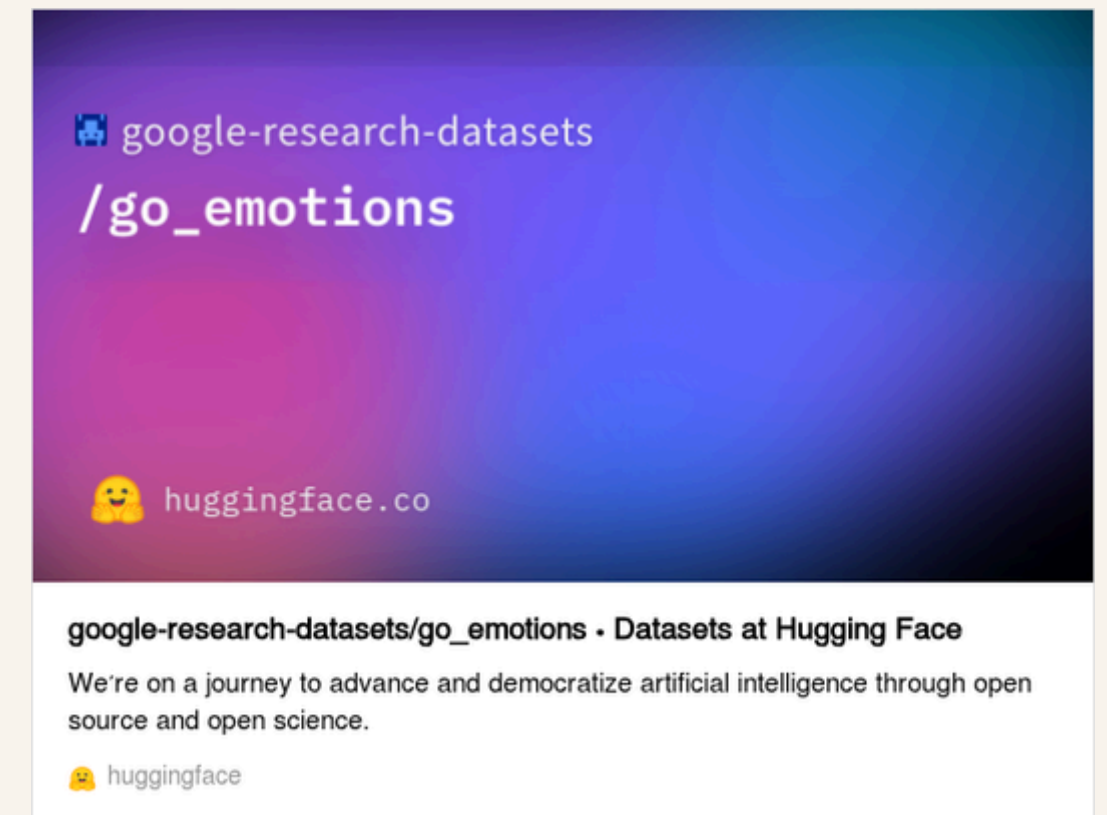
- No analysis of emotional trends across multiple journal entries (no longitudinal tracking)
 - Most systems use only a single modality (either text or speech)
 - Lack of personalized recommendations based on user interests and history
 - No connection between emotion detection and actionable outcomes (e.g., workload suggestions)
 - Privacy concerns due to use of raw sensitive data
- 

◆ **Novelty of the Project**

- Multimodal system integrating both text and speech inputs
 - Emotional trend tracking over time using a structured scoring approach
 - Personalized recommendations using user preferences, history, and feedback
 - Adaptive workload management based on emotional state
 - Privacy-preserving design using only processed emotional insights
 - Closed-loop system: emotion detection → analysis → actionable suggestions
- 

Dataset Details

- **Text data sourced from standard emotion datasets and preprocessed for training**
- **Multilingual support enabled through language detection and normalization**
- **Speech data processed using audio recordings with extracted acoustic features**
- **Journal inputs handled in two formats: text entries and voice recordings**



Methodology

Dataset Preparation

- Raw emotion datasets are cleaned and normalized
- Noise removal, text standardization, and formatting applied
- Ensures consistency across all training samples

Multimodal Input Processing

- User provides journal input as text or speech
- Speech is processed in two parallel paths:
 - Transcription → textual representation
 - Acoustic analysis → tone-based emotional signals

Emotion Detection & Filtering

- Text is cleaned and filtered to retain only writer-specific (first-person) content
- Emotion detection performed using XLM-RoBERTa (multilingual transformer)
- Multi-label emotion scores generated.
- Outputs:
 - Emotion probabilities
 - Dominant emotion
 - Confidence score

Methodology

Multimodal Fusion

- Combines:
 - Text-based emotion (XLM-RoBERTa)
 - Acoustic emotion (CNN)
- Confidence-based handling of conflicting signals
- Produces unified emotion vector per journal entry

Temporal Emotion Modeling

- Emotion vectors stored with timestamps in database
- Historical data used for time-series analysis
- Enables tracking of emotional progression across entries

Methodology

Agent-Based Decision Layer

- Agent 1: Emotional Momentum Score (EMS)
 - Computes intensity $\times$ frequency of negative emotions
 - Detects trends: stable / improving / escalating
- Agent 2: Burnout Risk Estimation
 - Combines sustained emotion, spikes, and emotional flatness
 - Outputs burnout risk score
- Agent 3: Workload Decision Engine
 - Time-aware logic (before/after evening)
 - Assigns actions (deep work / rest / recovery)
- Agent 4: Trigger Detection
 - Identifies recurring stress patterns from journal data

Personalization & Recommendation Generation

- Uses LLM-based generation
- Inputs: Emotional state, User interests, Feedback history, Detected triggers
- Generates personalized, scenario-aware suggestions

Closed Feedback Loop

- User feedback stored in database
- System adapts future recommendations
- Enables continuous personalization and improvement

Data Images

```
      text labels      id
0  My favourite food is anything I didn't have to...  [27]  eebbqej
1  Now if he does off himself, everyone will thin...  [27]  ed00q6i
2              WHY THE FUCK IS BAYLESS ISOING        [2]  eezlygj
3              To make her feel threatened          [14]  ed7ypvh
4              Dirty Southern Wankers                [3]  ed0bdzj
Index(['text', 'labels', 'id'], dtype='object')
```

Raw data

```
Final shape: (25721, 8)

      clean_text      mapped_emotions
2      why the fuck is bayless isoing      [anger]
3      to make her feel threatened        [fear]
5  omg peyton isn't good enough to help us in the...  [surprise]
6  yes i heard abt the f bombs! that has to be wh...  [trust]
7  we need more boards and to create a bit more s...  [optimism, anticipation]
```

Preprocessed Data

```
      mapped_emotions      label_vector \
0      ['trust']  [0.  1.  0.  0.  0.  0.  0.  0.  0.  0.]
1      ['trust']  [0.  1.  0.  0.  0.  0.  0.  0.  0.  0.]
2      ['trust']  [0.  1.  0.  0.  0.  0.  0.  0.  0.  0.]
3  ['anger', 'trust']  [0.  1.  0.  0.  0.  0.  1.  0.  0.  0.]
4  ['trust', 'anticipation']  [0.  1.  0.  0.  0.  0.  0.  1.  0.  0.]
```

```
      text_hi \
0  हां, यह 99.9 % निश्चित है, धन्यवाद, अपने सोने ...
1              ओह तुम मुझे मिल गया. अच्छा एक ।
2  आप समझ गए हैं कि आप पिछले 20 – जो भी दिन कर रह...
3  > मैं केवल उन विषयों पर बहस करूंगा जिनके बारे ...
4  धन्यवाद मुझे इसकी आवश्यकता है । मैं वर्तमान मे...
```

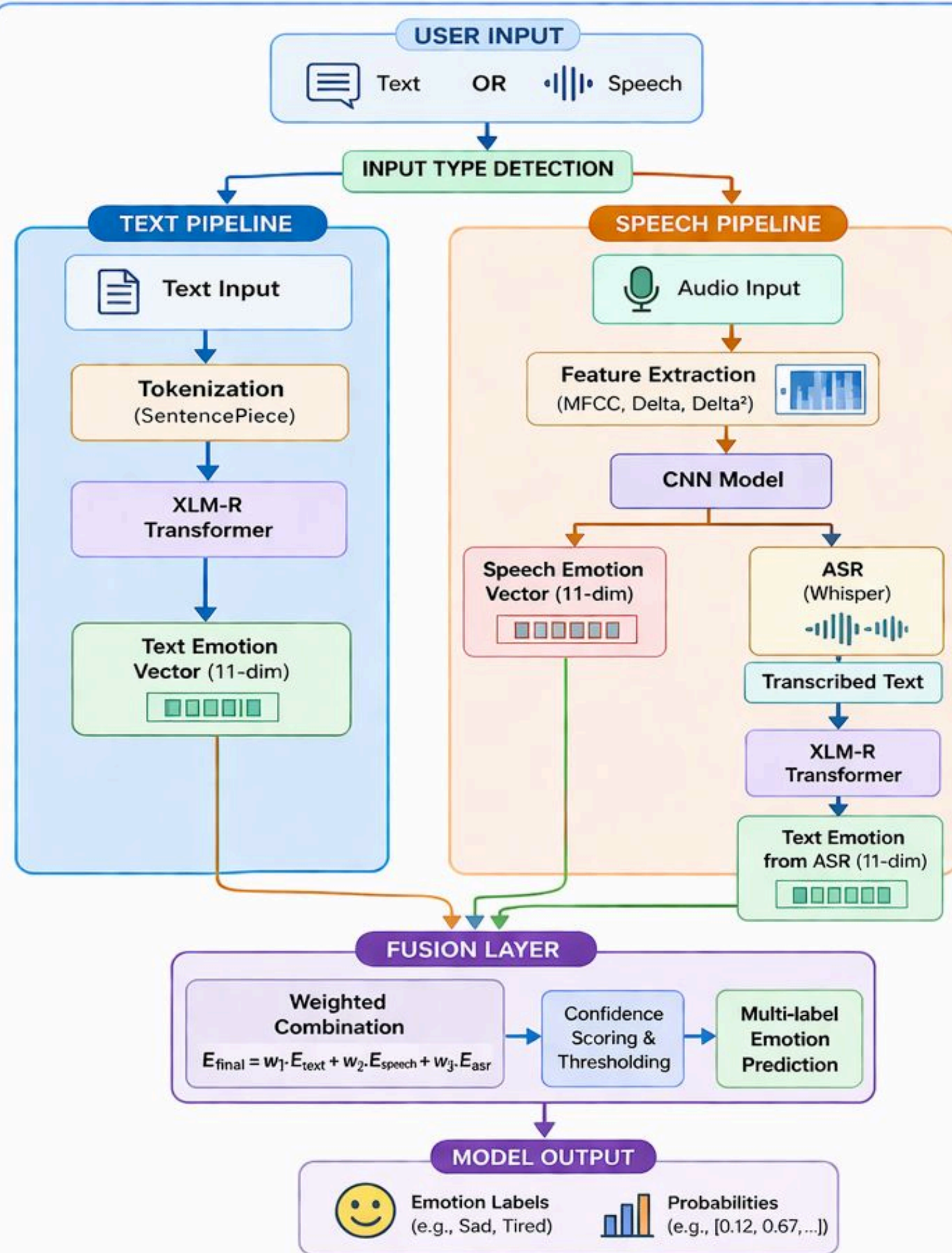
```
      text_te \
0  అవును ఇది 99.9 % ఖచ్చితంగా ఉంది ధన్యవాదాలు మీ ...
1              ఓహో, మీరు నన్ను పొందారు. మంచిది.
2  మీరు గత 20 – ఏ రోజులలో చేస్తున్నారో అదే చేస్తూ...
3  > నేను నాకు తెలిసిన విషయాలపై మాత్రమే చర్చిస్తా...
4  ధన్యవాదాలు నాకు ఇది అవసరం. నేను ప్రస్తుతం ఫై...
```

```
      text_ml
0  അതെ, ഇത് 99.9 % ഉറപ്പാണ്, നന്ദി, നിങ്ങളുടെ സ്വ...
1              ഓ, നിങ്ങൾക്ക് എന്നെ കിട്ടി. കൊള്ളാം.
2  കഴിഞ്ഞ 20 ദിവസമായി നിങ്ങൾ ചെയ്യുന്നതെന്തും ചെയ...
3  > എന്നിക്ക് അറിയാവുന്ന വിഷയങ്ങൾ മാത്രമേ ഞാൻ ചർച്ച...
4  നന്ദി, എന്നിക്ക് ഇത് ആവശ്യമാണ്. ഞാൻ ഇപ്പോൾ ഫ്രണ്ട്...
```

Multilingual Data

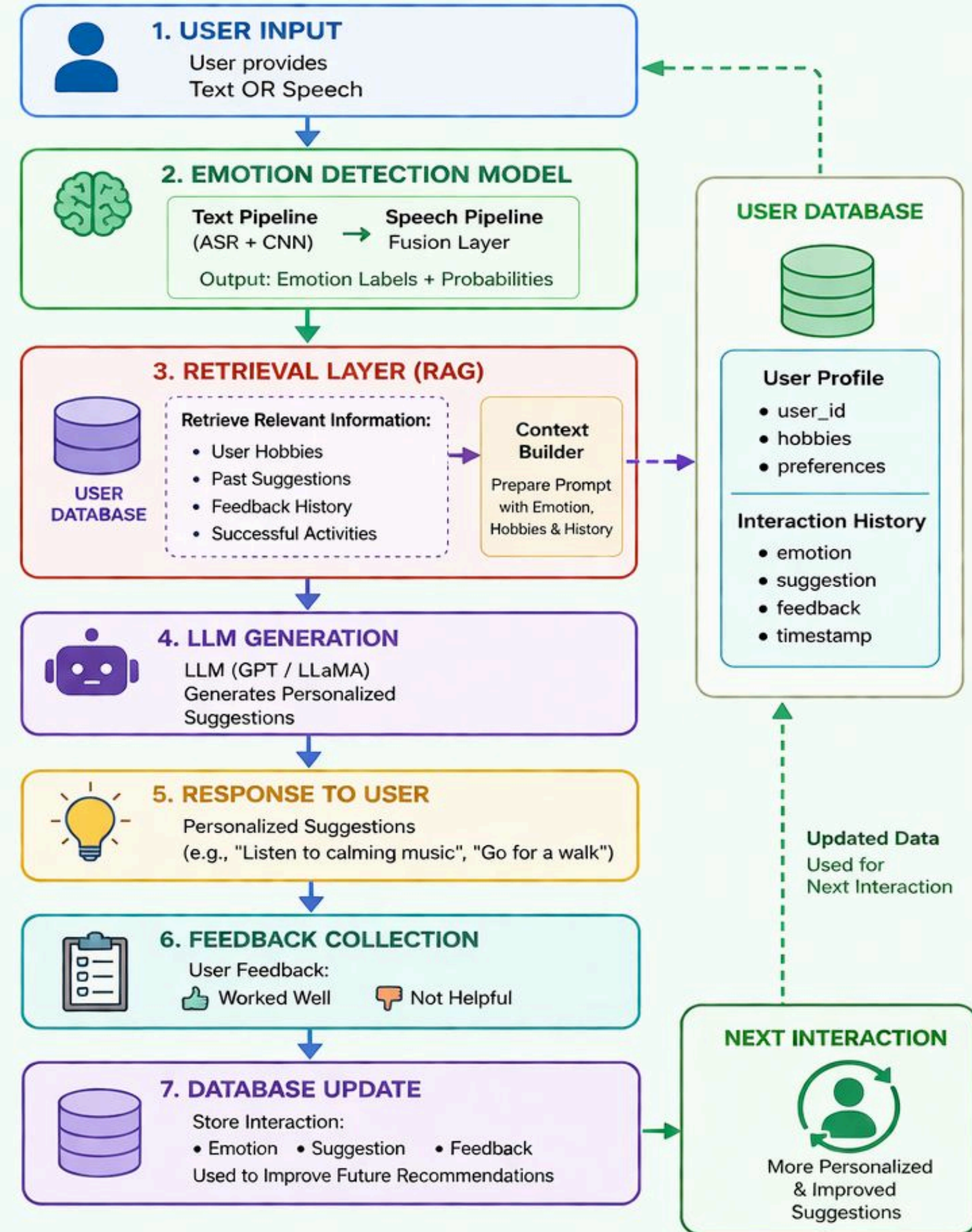
MODEL ARCHITECTURE

MULTIMODAL EMOTION DETECTION MODEL



SYSTEM ARCHITECTURE

EMOTION-AWARE RECOMMENDATION SYSTEM WITH RAG & FEEDBACK LOOP



Experimental Results

Training complete. Best Val Micro-F1: 0.6717

epoch	train_loss	val_loss	val_micro_f1	val_macro_f1
1	0.9534	0.7261	0.5944	0.5684
2	0.6769	0.6205	0.6359	0.6179
3	0.5550	0.5960	0.6590	0.6304
4	0.4616	0.6037	0.6630	0.6382
5	0.3944	0.6197	0.6717	0.6471

```
-----
Text:      daddy drove but he was kinda feared. I had a really good time ove
Dominant: joy
Active:     ['joy']
Top 3:      [('joy', 0.847), ('trust', 0.513), ('love', 0.304)]
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```

```
Text:      Finally submitted my project. Feeling so proud and relieved.
Dominant: joy
Active:     ['joy']
Top 3:      [('joy', 0.991), ('trust', 0.384), ('surprise', 0.073)]
```

- Model performs well on frequent and clearly expressed emotions
- Lower scores in certain classes indicate:
 - Class imbalance
 - Subtle emotional expressions
- Multi-label predictions successfully capture co-existing emotions

Test Loss: 0.6752

Test Micro-F1: 0.6603

Test Macro-F1: 0.6326

Per-class report:

	precision	recall	f1-score	support
joy	0.73	0.68	0.70	656
trust	0.63	0.69	0.66	452
fear	0.55	0.74	0.63	117
surprise	0.66	0.68	0.67	571
sadness	0.58	0.84	0.69	304
disgust	0.34	0.62	0.43	104
anger	0.64	0.64	0.64	659
anticipation	0.55	0.71	0.62	397
love	0.74	0.83	0.78	536
optimism	0.65	0.79	0.71	335
pessimism	0.34	0.53	0.41	204
micro avg	0.62	0.71	0.66	4335
macro avg	0.58	0.70	0.63	4335
weighted avg	0.63	0.71	0.66	4335
samples avg	0.66	0.73	0.67	4335

Natural Language Processing



Thank You!

